

SUMMER TRAINING/INTERNSHIP

PROJECT REPORT

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**Customer Churn Prediction in Subscription-Based Services
using Machine Learning**

Submitted by

NAME	REGISTRATION NUMBER
SUARNAM LAKSHMI	12400532
MALLESH	
PUPPALA UDAY KIRAN	12505113

Course Code: PETV149

Under the guidance of

Ms. Sandeep Kaur-UID: 23614

Ms. Savleen Kaur-UID: 18306

School Of Computer Science Engineering



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This is to certify that Mr./Ms. Suvarnam Lakshmi Mallesh S/D/W/o Mr. Suvarnam Vijay Kumar
student of School of AI and Emerging Technologies Registration No. 12400532
pursuing Bachelor of Technology (Computer Science and Engineering) completed
skill development course named Next-Gen Data Bootcamp : SQL, PowerBI, Python, ML & AI Automation
organized by Centre for Professional Enhancement Lovely Professional University
from 11-06-2026 to 24-07-2026 and obtained O Grade.

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Prepared by
(Administrative Officer-Records)


Programme Coordinator
Centre for Professional Enhancement


Head of School
School of AI and Emerging Technologies

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Abstract

Customer churn — the loss of subscribers to a service — is one of the most pressing challenges faced by subscription-based businesses, particularly in the telecommunications sector, where acquiring a new customer is significantly more expensive than retaining an existing one. This project develops an end-to-end machine learning pipeline to predict customer churn using the IBM Telco Customer Churn dataset, which contains demographic, account, and service-usage information for 7,043 customers across 21 attributes.

The workflow covers data cleaning, exploratory data analysis (EDA), feature engineering, encoding, and the training of three classification algorithms — Logistic Regression, Decision Tree, and Random Forest. Model performance was evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC, since the target variable is imbalanced (73.46% non-churn vs. 26.54% churn). Logistic Regression achieved the best overall balance across metrics and was further optimised through Grid Search CV, achieving a tuned Accuracy of 79.8% and an ROC-AUC of 0.721 on the test set. Statistical validation using the stats models library and Variance Inflation Factor (VIF) analysis confirmed the significance of key predictors such as tenure, Contract type, Phone Service, and Online Security. The final model was integrated with a random and manual customer prediction module that outputs churn probability, and the analytical insights were consolidated into an interactive Power BI dashboard for business stakeholders. The project concludes with actionable business recommendations aimed at reducing churn through contract incentives, improved onboarding, and targeted retention offers.

Chapter 1: Introduction

1.1 Background

Subscription-based businesses, particularly telecommunications providers, depend heavily on long-term customer relationships for sustained revenue. Customer churn — when a subscriber discontinues a service — directly affects profitability, and industry studies consistently show that retaining an existing customer is far more cost-effective than acquiring a new one. Understanding the factors that drive churn and identifying at-risk customers in advance allows businesses to intervene proactively rather than reactively.

1.2 Problem Statement

Develop a machine learning model to predict customer churn in subscription-based services by analysing customer demographics, service usage, billing information, and contract details. The model should identify customers who are likely to discontinue the service so that businesses can take proactive customer retention measures.

1.3 Objectives

- To clean and preprocess the Telco Customer Churn dataset, correcting data types and handling missing values.
- To perform exploratory data analysis to uncover patterns and relationships between customer attributes and churn behaviour.
- To engineer meaningful derived features that better capture customer engagement and value.
- To build, train, and compare Logistic Regression, Decision Tree, and Random Forest classifiers.
- To tune the best-performing model using GridSearchCV and validate it statistically.
- To develop a prediction module capable of scoring individual customers with churn probability.
- To visualise key findings and model outputs through an interactive Power BI dashboard.

- To translate analytical findings into actionable business recommendations for customer retention.

1.4 Scope

The scope of this project covers the complete data science lifecycle applied to a single, publicly available dataset: data acquisition, cleaning, exploratory analysis, feature engineering, supervised classification modelling, statistical validation, and business intelligence reporting. The project is limited to the IBM Telco Customer Churn dataset and does not incorporate real-time data streams or external data sources. The trained model is intended as a decision-support tool for retention teams rather than a fully automated intervention system.

Chapter 2: Dataset Description

The project uses the IBM Telco Customer Churn dataset, a widely used benchmark dataset in customer analytics. It was loaded from the file `WA_Fn-UseC_-Telco-Customer-Churn.csv` and contains 7,043 customer records described by 21 attributes, combining demographic details, subscribed services, account/billing information, and the churn label.

2.1 Feature Summary

Category	Attributes
Demographics	gender, SeniorCitizen, Partner, Dependents
Account Information	tenure, Contract, PaperlessBilling, PaymentMethod
Services Subscribed	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Billing	MonthlyCharges, TotalCharges
Target Variable	Churn (Yes / No)

2.2 Dataset Snapshot

A preliminary inspection of the dataset (`df.shape`) confirmed 7,043 rows and 21 columns, with `df.info()` showing 18 categorical (object) columns, 2 integer columns (`SeniorCitizen`, `tenure`), and 1 float column (`MonthlyCharges`) prior to type correction.

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	No	Month-to-month
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No	No	One year
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	No	No	Month-to-month
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No	No	One year
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	No	No	Month-to-month

PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
No	No phone service	DSL	No	...	No	No	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No
Yes	No	DSL	Yes	...	Yes	No	No	No	One year	No	Mailed check	56.95	1889.5	No
Yes	No	DSL	Yes	...	No	No	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes
No	No phone service	DSL	Yes	...	Yes	Yes	No	No	One year	No	Bank transfer (automatic)	42.30	1840.75	No
Yes	No	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes	Electronic check	70.70	151.65	Yes

Figure 2.1: Dataset Overview (*df.head()* output)

Chapter 3: Technologies and Requirements

3.1 Technologies Used

Category	Tools / Libraries
Programming Language	Python
Data Manipulation	Pandas, NumPy
Data Visualisation	Matplotlib, Seaborn
Machine Learning	Scikit-learn (LogisticRegression, DecisionTreeClassifier, RandomForestClassifier, GridSearchCV)
Statistical Analysis	Statsmodels (Logit, Variance Inflation Factor)
Model Persistence	Joblib
Business Intelligence	Microsoft Power BI, Power Query
Development Environment	Jupyter Notebook

3.2 Software Requirements

- Python 3.x with Jupyter Notebook / JupyterLab
- Required libraries: pandas, numpy, matplotlib, seaborn, scikit-learn, statsmodels, joblib
- Microsoft Power BI Desktop for dashboard development
- Operating System: Windows / macOS / Linux (cross-platform compatible)

3.3 Hardware Requirements

- Processor: Intel i3 / AMD equivalent or higher
- RAM: 8 GB minimum (recommended for smooth Power BI rendering)
- Storage: 2 GB free disk space for dataset, notebook, and model artifacts

Chapter 4: Methodology

The project follows a structured, sequential machine learning workflow: Data Collection → Data Cleaning → Exploratory Data Analysis → Feature Engineering → Encoding → Train-Test Split → Model Training → Model Evaluation → Hyperparameter Tuning → Final Logistic Regression Model → Random Customer Prediction → Business Recommendation. Each stage builds on the outputs of the previous one, ensuring that the final model is both statistically sound and practically deployable.

4.1 Data Collection

The dataset was loaded directly into a Pandas DataFrame using `pd.read_csv()`. Initial checks — `df.shape`, `df.columns`, `df.info()`, and `df.describe()` — were performed to understand the structure, data types, and statistical distribution of the 21 features before any transformation was applied.

4.2 Data Cleaning and Preprocessing

4.2.1 Data Type Correction

The Total Charges column, despite representing a monetary amount, was stored as an object (string) type rather than numeric — a common artefact of blank spaces or non-numeric placeholder values in the raw data. Since machine learning algorithms require numeric input, this column was converted using `pd.to_numeric(df['TotalCharges'], errors='coerce')`, which coerces invalid entries to NaN for downstream handling.

4.2.2 Handling Missing Values

The type conversion revealed 11 missing values in Total Charges. Investigation showed that all 11 affected records had a tenure of 0 months — these were newly joined customers who had not yet been billed, so a Total Charges value did not logically exist for them. Since the records were otherwise valid and informative, the missing values were imputed with 0 using `fillna(0, inplace=True)` rather than being dropped.

4.2.3 Removing Unnecessary Columns

The customerID column is a unique identifier with no predictive value for churn behaviour. It was removed from the dataset using `df.drop('customerID', axis=1, inplace=True)` prior to modelling.

4.2.4 Duplicate Records

After removing customerID, 22 duplicate rows were detected. These correspond to different customers who happen to share identical demographic and service attribute combinations — once the unique identifier was dropped, they became indistinguishable. Since each row still represents valid customer information, the duplicates were retained rather than removed.

4.3 Exploratory Data Analysis (EDA)

EDA was conducted to uncover patterns, relationships, and factors influencing churn before feature engineering and modelling.

4.3.1 Target Variable Distribution

The Churn distribution is imbalanced: 73.46% of customers stayed (No) and 26.54% churned (Yes). Because of this imbalance, evaluation was based on Precision, Recall, F1-Score, and ROC-AUC in addition to Accuracy, rather than Accuracy alone.

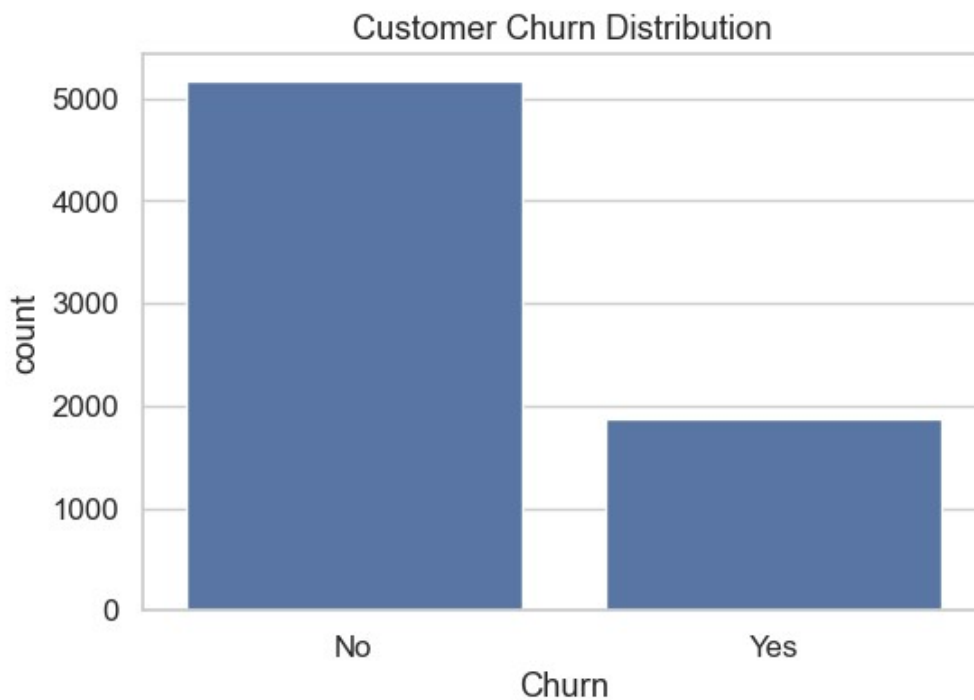


Figure 4.1: Customer Churn Distribution

4.3.2 Numerical Feature Summary

The average customer tenure is 32.37 months, ranging from 0 to 72 months. The average monthly charge is ₹64.76, while the average total charge is ₹2,279.73, reflecting substantial variation in customer spending accumulated over different subscription lengths.

4.3.3 Distribution of Customer Tenure

The tenure distribution is bimodal — a large number of customers are either new (0–5 months) or long-term (around 70–72 months), suggesting that retention dynamics differ sharply between newly acquired and loyal customer segments, making tenure a valuable predictive feature.

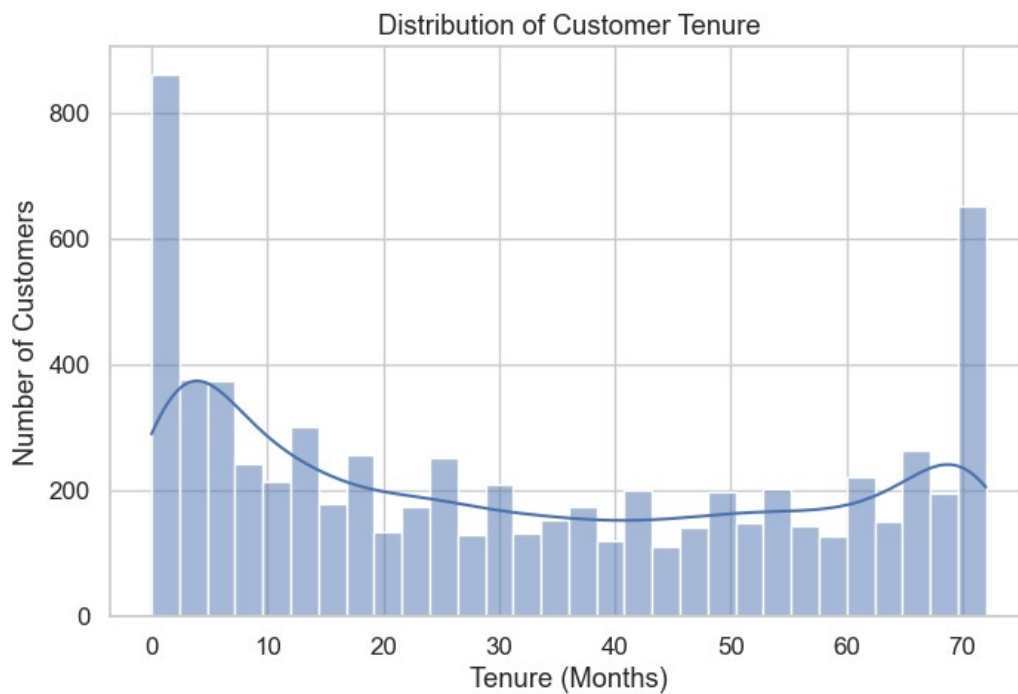


Figure 4.2: Distribution of Customer Tenure

4.3.4 Distribution of Monthly and Total Charges

Monthly charges range from ₹18.25 to ₹118.75, with visible clusters around lower (~₹20) and higher (₹70–₹100) price points corresponding to different subscription plans. Total Charges is positively skewed, since customers with longer tenure naturally accumulate higher totals.

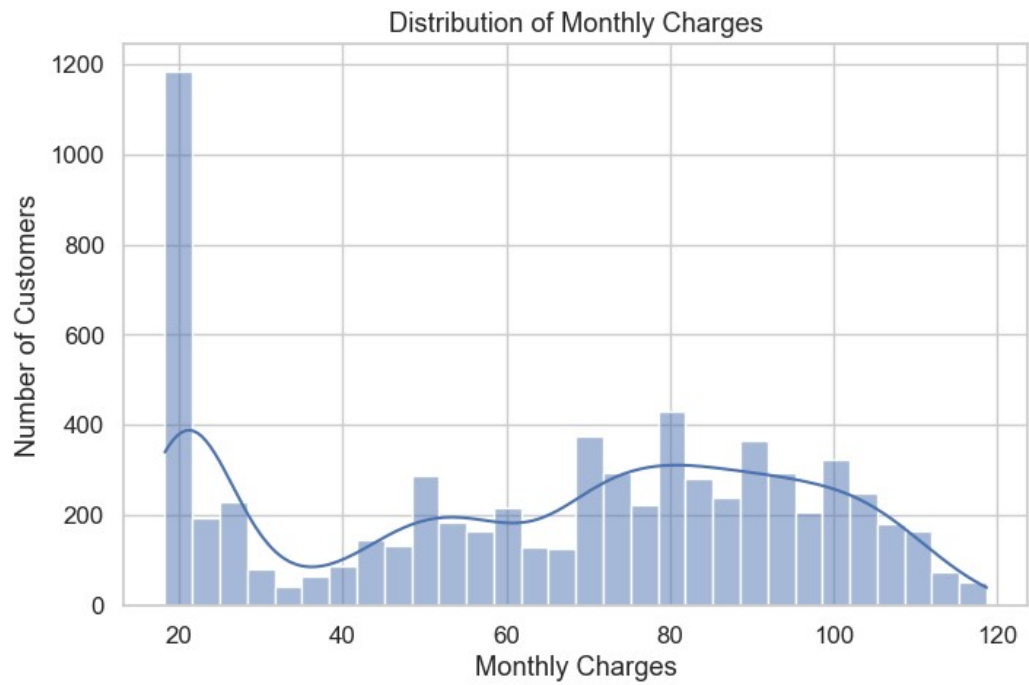


Figure 4.3: Distribution of Monthly Charges

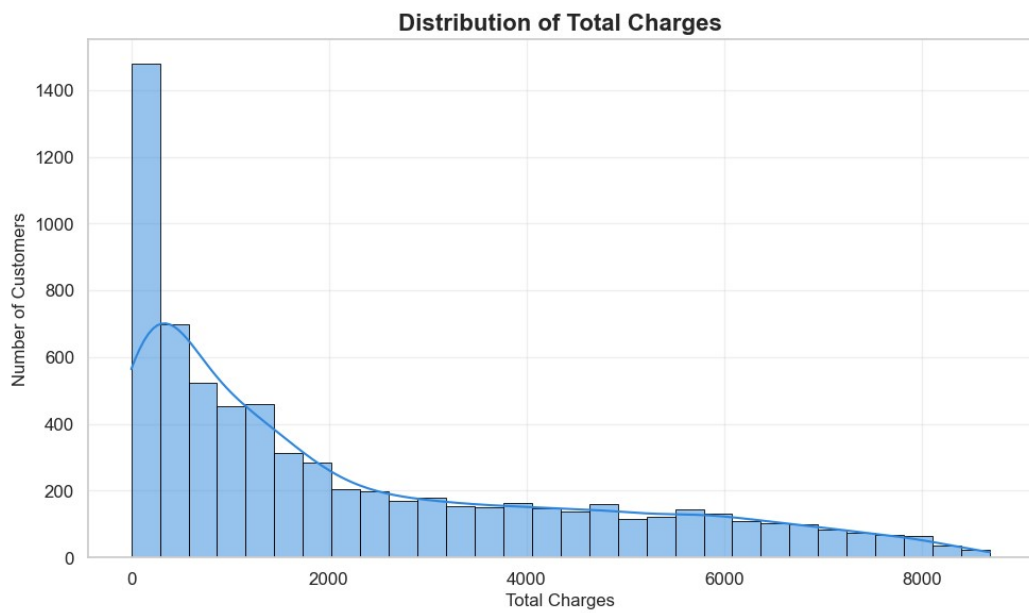


Figure 4.4: Distribution of Total Charges

4.3.5 Bivariate Analysis: Tenure, Charges, and Churn

Customers who churn generally have shorter tenure than those who stay, indicating that newer customers are more likely to leave. Customers with higher monthly charges churn more frequently, while customers who stayed accumulated significantly higher total charges over time than those who churned — consistent with longer-tenured customers contributing more revenue before any potential churn event.

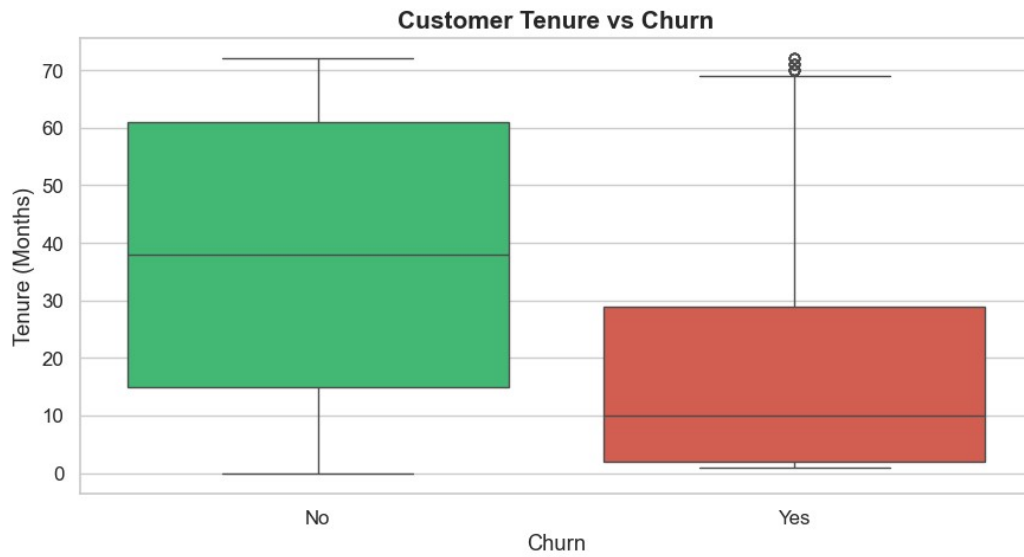


Figure 4.5: Customer Tenure vs Churn

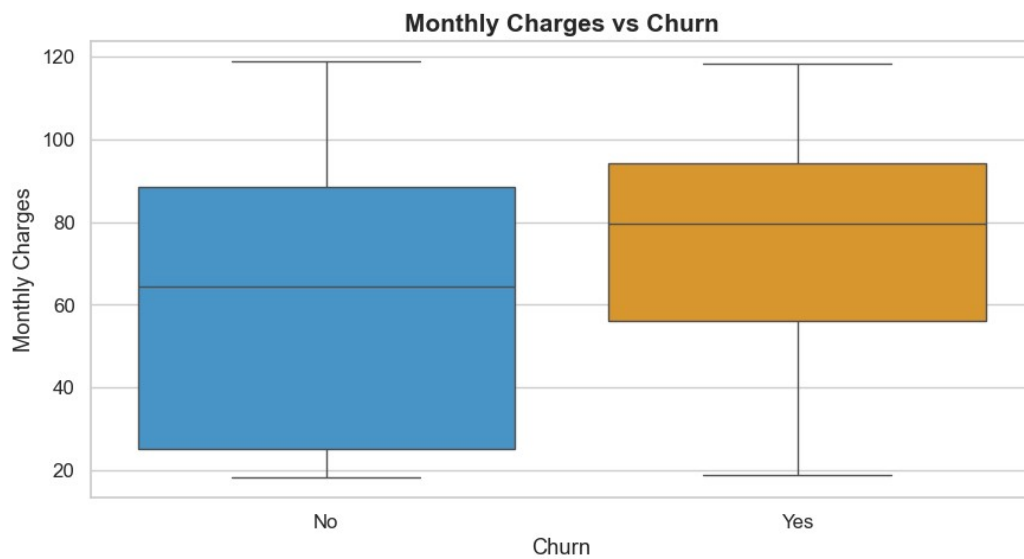


Figure 4.6: Monthly Charges vs Churn

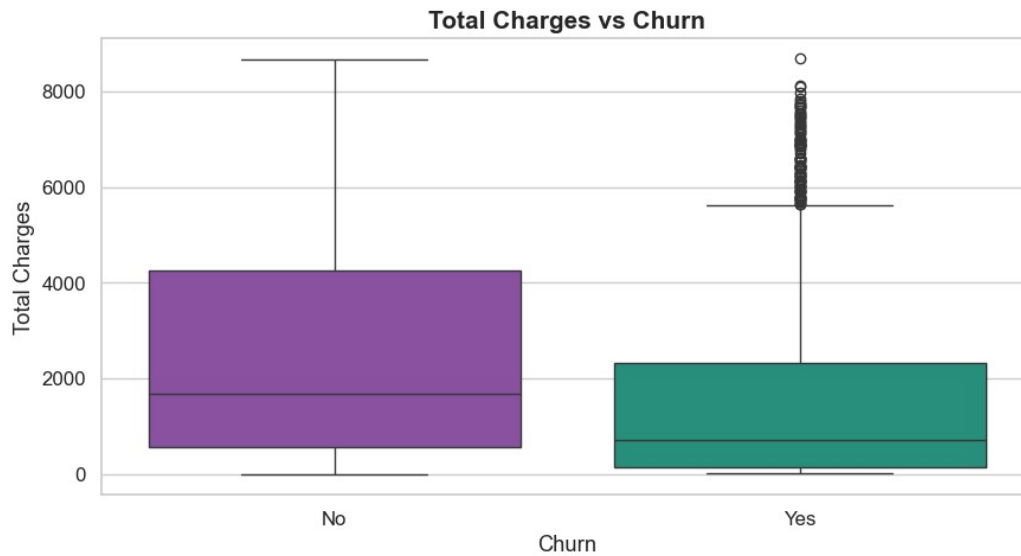


Figure 4.7: Total Charges vs Churn

4.3.6 Contract Type, Payment Method, and Internet Service vs Churn

Customers on Month-to-Month contracts show the highest churn rate, while One-Year and Two-Year contract holders are markedly more likely to remain — indicating that longer commitment periods improve retention. Customers paying via Electronic Check churn the most, whereas Automatic Bank Transfer and Automatic Credit Card payments are associated with the lowest churn, suggesting that automatic payment methods support retention. Internet service type was also examined for its relationship with churn percentage.

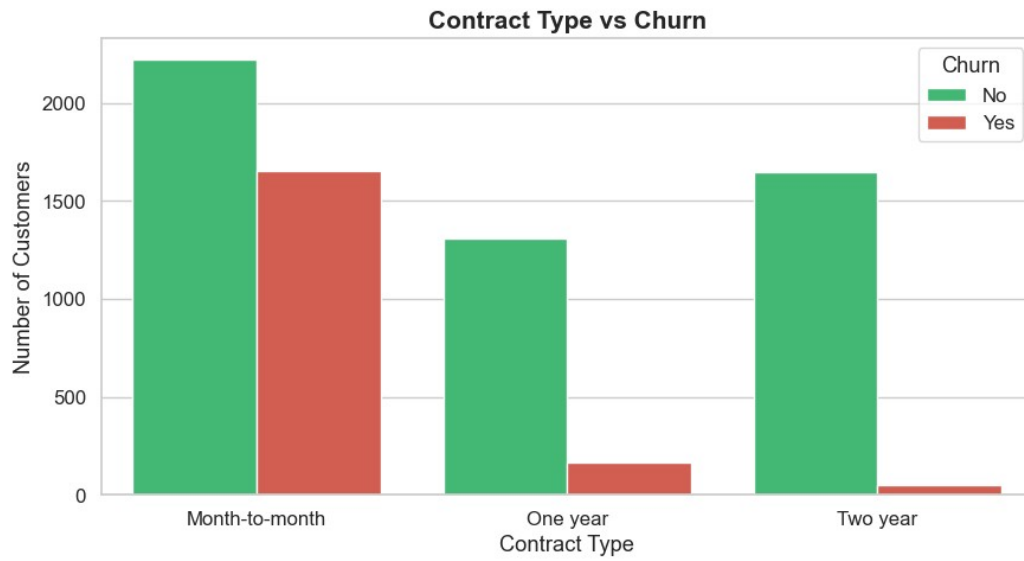


Figure 4.8: Contract Type vs Customer Churn

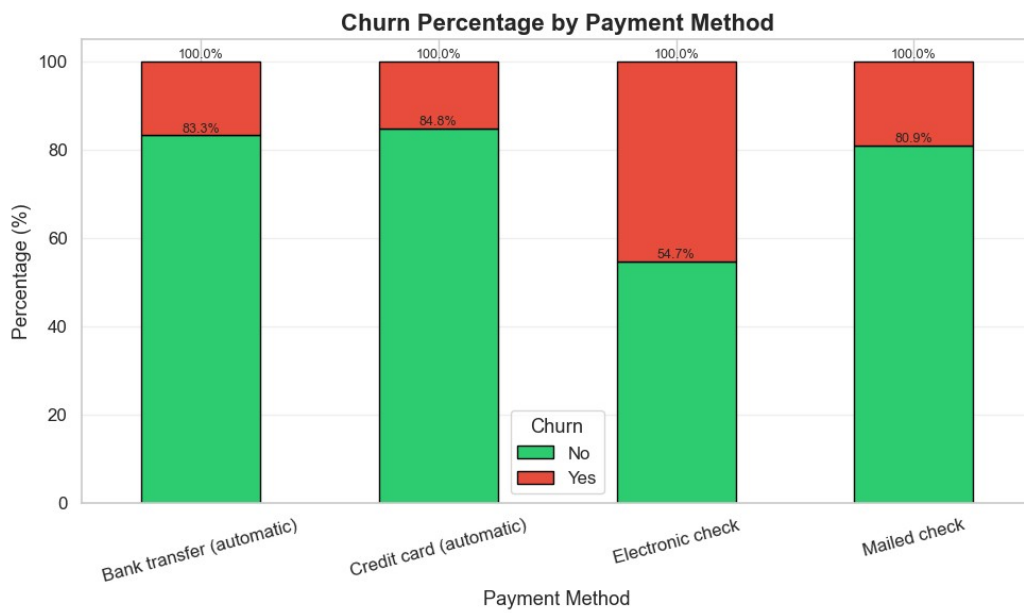


Figure 4.9: Payment Method vs Customer Churn

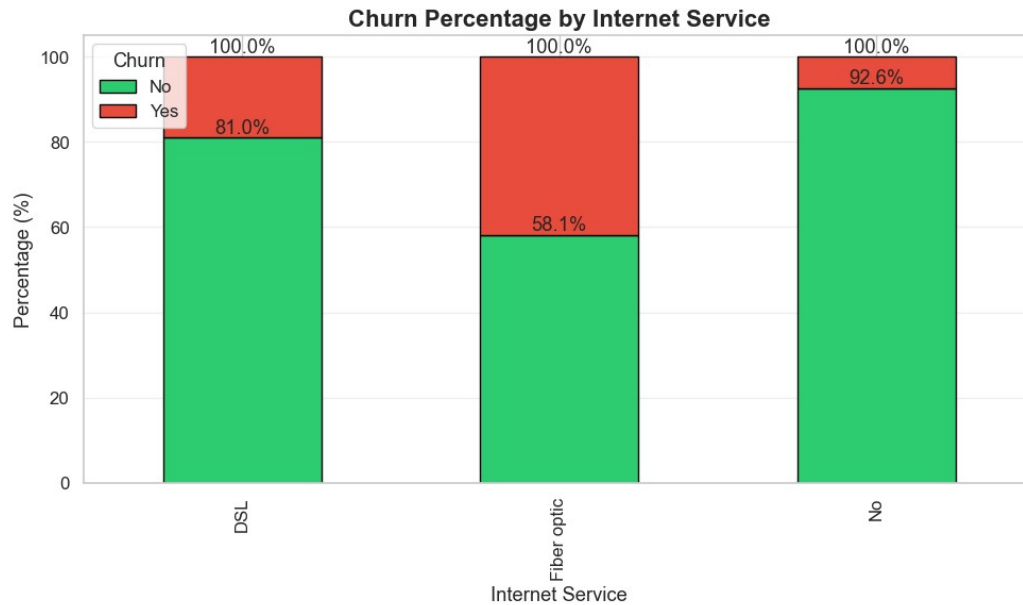


Figure 4.10: Internet Service vs Customer Churn

4.4 Feature Engineering

Five derived features were engineered from the cleaned dataset to better capture customer engagement, spending behaviour, and value to the business. Each feature is designed to expose a pattern that raw columns alone do not directly represent.

4.4.1 Total Services

Purpose: customers using more services tend to be more engaged with the company and may exhibit different churn behaviour than those with minimal service adoption. Total Services counts, for each customer, how many of the nine service-related columns (Phone Service, Multiple Lines, Internet Service, Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, Streaming Movies) are set to an active value ('Yes', 'DSL', or 'Fiber optic').

4.4.2 Avg Monthly Spend

Purpose: average monthly spending gives a normalised view of a customer's financial value regardless of tenure length. It is calculated as Total Charges divided by tenure, with a safeguard (via np.where) that assigns 0 for customers with zero tenure to avoid division errors.

4.4.3 Customer Lifetime Value

Purpose: customers with higher lifetime value contribute more revenue and are more valuable to retain, making this a useful feature for prioritising retention effort. It is estimated as the product of Monthly Charges and tenure.

4.4.4 Long Term Customer

Purpose: customers who have remained subscribed for longer periods generally carry lower churn risk. This binary flag classifies customers with tenure of 24 months or more as long-term customers (1), and all others as 0.

4.4.5 HighValueCustomer

Purpose: customers paying above-average monthly charges represent higher revenue contribution and therefore warrant closer retention attention. This binary flag identifies customers whose Monthly Charges exceed the 75th percentile of the distribution as high-value (1), and all others as 0.

4.5 Data Encoding

All categorical (object-type) columns in the feature matrix X were converted into numeric form using Scikit-learn's Label Encoder, with a separate encoder fitted per column and the resulting category-to-code mappings stored in a dictionary for later interpretation. The target variable Churn was similarly encoded ('No' → 0, 'Yes' → 1) using a dedicated Label Encoder . After encoding, the feature matrix comprised 24 fully numeric columns (20 integer and 4 float), ready for model training.

Column	Category	Encoded Value
gender	Female / Male	0 / 1
Partner	No / Yes	0 / 1
Dependents	No / Yes	0 / 1
PhoneService	No / Yes	0 / 1
MultipleLines	No / No phone service / Yes	0 / 1 / 2
InternetService	DSL / Fiber optic / No	0 / 1 / 2
OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies	No / No internet service / Yes	0 / 1 / 2
Contract	Month-to-month / One year / Two year	0 / 1 / 2
PaperlessBilling	No / Yes	0 / 1
PaymentMethod	Bank transfer (automatic) / Credit card (automatic) / Electronic check / Mailed check	0 / 1 / 2 / 3

4.6 Train-Test Split

The dataset was split into training and testing subsets using an 80:20 ratio with `train_test_split(X, y, test_size=0.20, random_state=42, stratify=y)`. Stratification on the target variable preserved the original 73.46% / 26.54% class ratio in both subsets, resulting in a training set of 5,634 records and a test set of 1,409 records, each with 24 features.

Chapter 5: Model Building

Three supervised classification algorithms were trained on the encoded training data and evaluated on the held-out test set to identify the most suitable model for churn prediction.

5.1 Logistic Regression (Baseline)

A Logistic Regression model (`max_iter=10000`, `random_state=42`) was trained as the baseline classifier. On the test set, it achieved an Accuracy of 79.56%, Precision of 0.634, Recall of 0.543, F1-Score of 0.585, and ROC-AUC of 0.715 for the churn class.

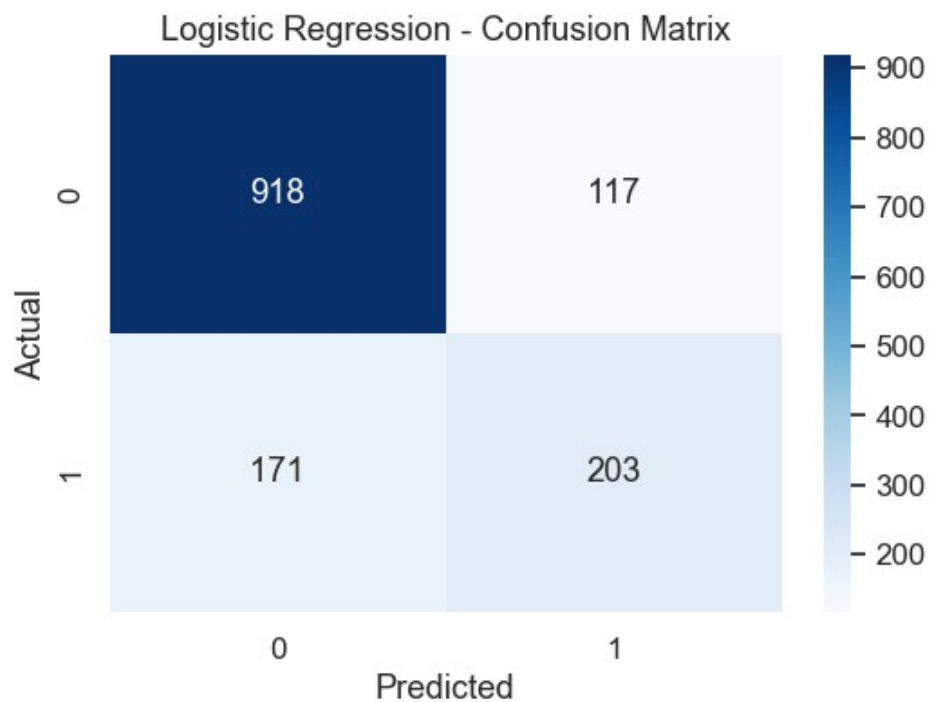


Figure 5.1: Logistic Regression – Confusion Matrix

5.2 Decision Tree Classifier

A Decision Tree Classifier (`random_state=42`) was trained with default parameters. It achieved a lower Accuracy of 72.04%, Precision of 0.474, Recall of 0.497, F1-Score of 0.486, and ROC-AUC of 0.649 — indicating a tendency to overfit the training data and generalise less effectively than Logistic Regression.

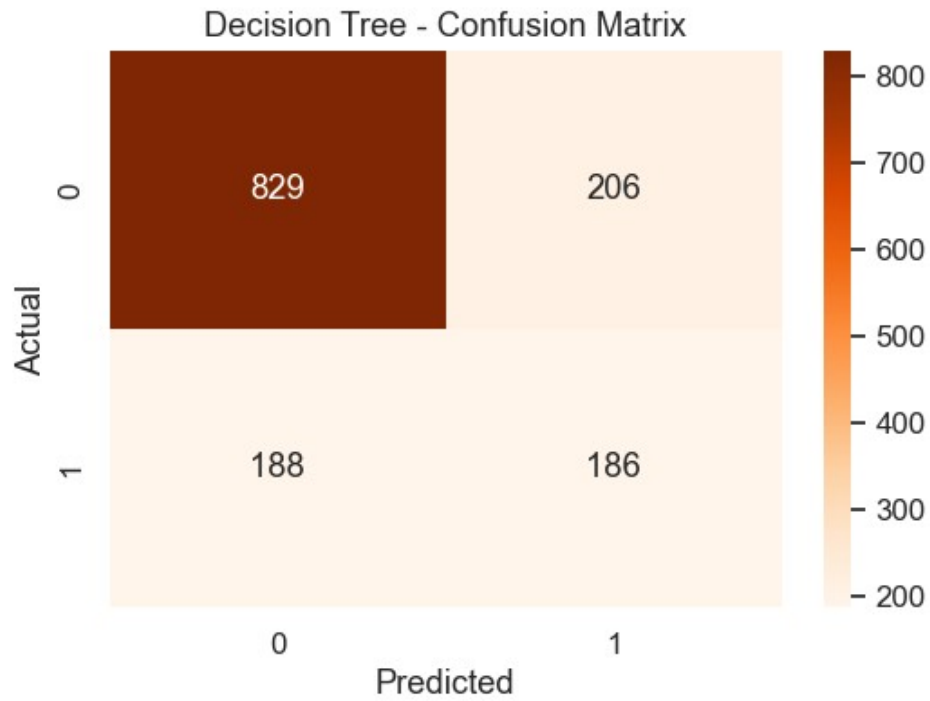


Figure 5.2: Decision Tree – Confusion Matrix

5.3 Random Forest Classifier

A Random Forest Classifier (`n_estimators=200`, `random_state=42`) was trained as an ensemble alternative. It achieved an Accuracy of 78.50%, Precision of 0.621, Recall of 0.487, F1-Score of 0.546, and ROC-AUC of 0.690 — an improvement over the single Decision Tree but still short of Logistic Regression on F1-Score and ROC-AUC.

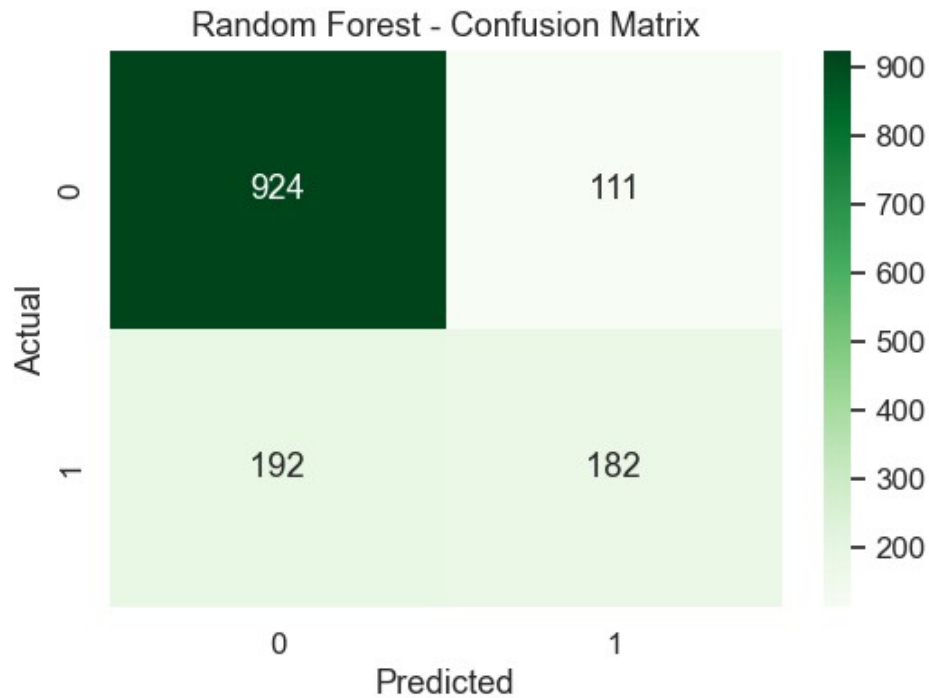


Figure 5.3: Random Forest – Confusion Matrix

5.4 Model Comparison

The three models were compared side by side across all evaluation metrics on the same test set:

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.796	0.634	0.543	0.585	0.715
Decision Tree	0.720	0.474	0.497	0.486	0.649
Random Forest	0.785	0.621	0.487	0.546	0.690

Logistic Regression achieved the highest Accuracy, F1-Score, and ROC-AUC among the three models. It was therefore selected as the final candidate model for customer churn prediction and carried forward for hyperparameter tuning.

5.5 Hyperparameter Tuning

Grid Search CV was used to identify the optimal hyperparameters for Logistic Regression, searching over a regularisation strength grid $C \in \{0.01, 0.1, 1, 10\}$ and solver $\in \{\text{liblinear}, \text{lbfgs}\}$, with 5-fold cross-validation optimised for F1-score. The

search identified $C = 0.1$ with the liblinear solver as the best configuration, achieving a cross-validated F1-score of 0.599.

5.6 Final Model Selection

The tuned Logistic Regression model (`best_lr`) was evaluated on the test set, achieving an Accuracy of 79.84%, Precision of 0.638, Recall of 0.556, F1-Score of 0.594, and ROC-AUC of 0.721 — a measurable improvement over the untuned baseline across every metric, particularly Recall and F1-Score, which matter most for identifying at-risk customers. This tuned model was saved for deployment using `joblib.dump(best_lr, 'customer_churn_model.pkl')`.

Chapter 6: Model Evaluation

6.1 Classification Report (Final Tuned Model)

Class	Precision	Recall	F1-Score	Support
0 (No Churn)	0.85	0.89	0.87	1035
1 (Churn)	0.64	0.56	0.59	374
Accuracy			0.80	1409
Macro Avg	0.74	0.72	0.73	1409
Weighted Avg	0.79	0.80	0.79	1409

6.2 Confusion Matrix Analysis

The confusion matrix for the tuned model confirms strong performance on the majority (No Churn) class, with a Precision of 0.85 and Recall of 0.89. For the minority Churn class, Precision improved to 0.64 while Recall reached 0.56 — meaning the model correctly identifies 56% of customers who actually churn. This gap between the two classes is a direct consequence of the class imbalance in the underlying data (73.46% vs. 26.54%) and represents the primary area for future improvement.

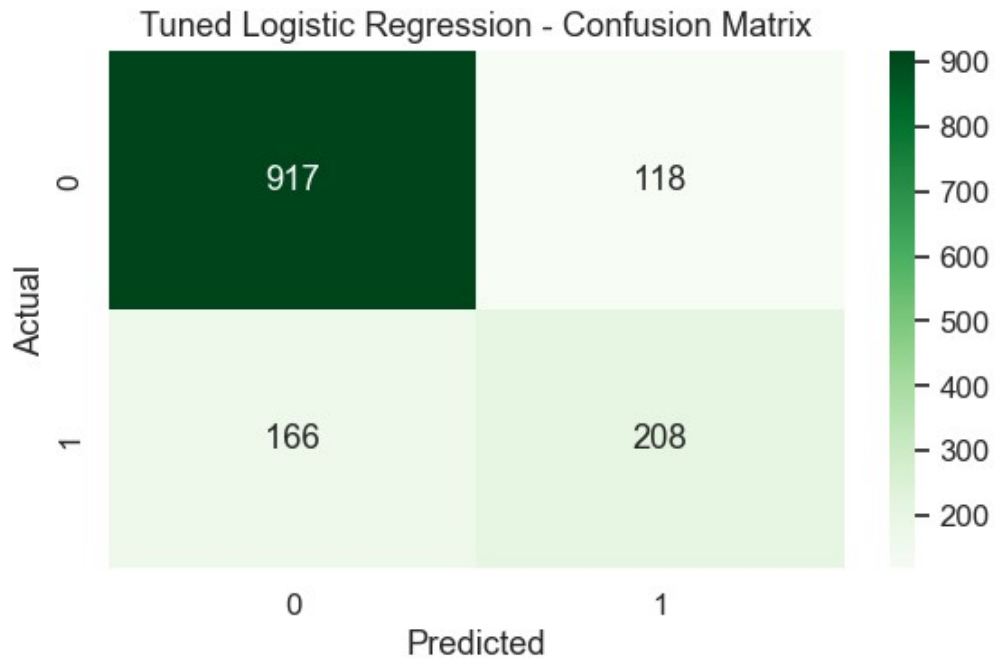


Figure 6.1: Tuned Logistic Regression – Confusion Matrix

6.3 ROC Curve

The ROC Curve evaluates the model's ability to distinguish between churn and non-churn customers by plotting the True Positive Rate against the False Positive Rate across all classification thresholds. The final model achieved an Area Under the Curve (AUC) consistent with the reported ROC-AUC of 0.721, indicating a moderate-to-good ability to discriminate between the two classes — well above the 0.5 baseline of random guessing.

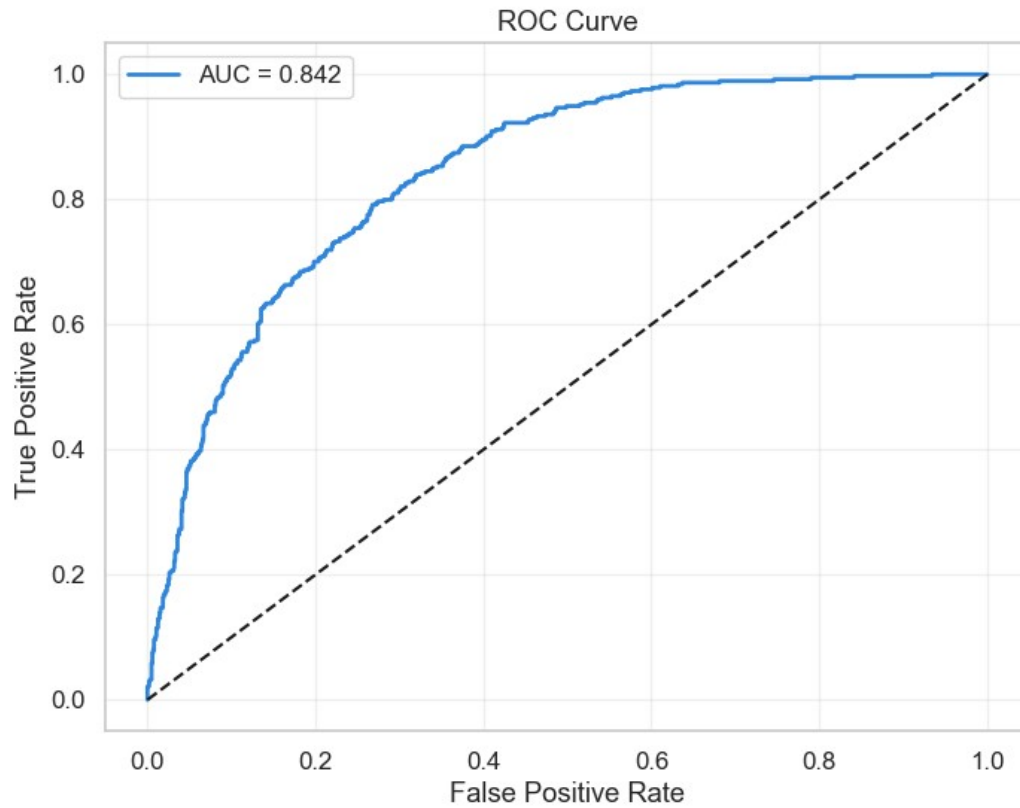


Figure 6.2: ROC Curve for the Tuned Logistic Regression Model

6.4 Feature Importance

Feature importance was derived from the absolute value of the Logistic Regression coefficients. The ten most influential features are:

Feature	Coefficient	Direction of Influence
PhoneService	-0.751	Reduces churn likelihood
Contract	-0.648	Reduces churn likelihood (longer contracts)
PaperlessBilling	+0.337	Increases churn likelihood
OnlineSecurity	-0.235	Reduces churn likelihood
Dependents	-0.230	Reduces churn likelihood

Feature	Coefficient	Direction of Influence
TechSupport	-0.217	Reduces churn likelihood
TotalServices	-0.170	Reduces churn likelihood
MultipleLines	+0.156	Increases churn likelihood
SeniorCitizen	+0.155	Increases churn likelihood
HighValueCustomer	+0.093	Increases churn likelihood

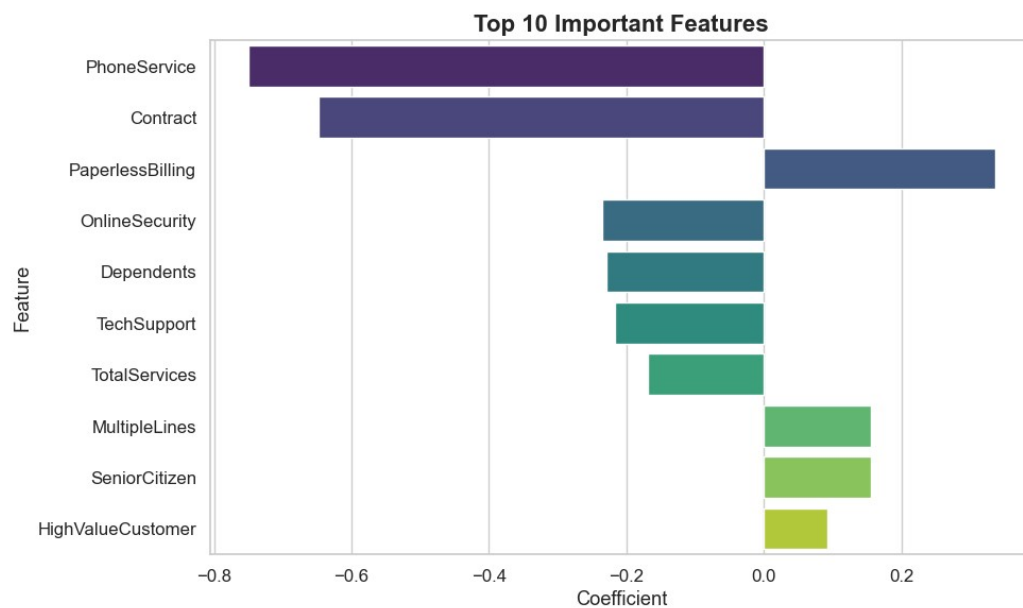


Figure 6.3: Top 10 Important Features (Logistic Regression Coefficients)

6.5 Statistical Validation using Stats models

To statistically validate the relationships identified by the Logistic Regression model, a parallel Logit model was fitted using the stats models library, which reports coefficients, standard errors, z-statistics, p-values, and confidence intervals. The model converged successfully with a Pseudo R-squared of 0.282 (Log-Likelihood = -2925.7 vs. LL-Null = -4075.1), confirming that the selected features jointly explain a meaningful share of the variation in churn. Predictors such as tenure, Phone Service, Internet Service, and Online Security were statistically significant at the 5% level ($p < 0.05$), while gender and Partner were not significant predictors of churn.

6.6 Multicollinearity Check (VIF)

Variance Inflation Factor (VIF) analysis was performed to detect multicollinearity among the independent variables. Several engineered and billing-related features — Total Services, Customer Lifetime Value, Total Charges, and Monthly Charges — showed very high VIF values (all exceeding 900), indicating strong linear dependence among these variables, largely because Customer Lifetime Value and Avg Monthly Spend are mathematically derived from Monthly Charges, tenure, and Total Charges. In contrast, most original categorical predictors (e.g., Contract, Paperless Billing, Payment Method, Senior Citizen, gender) showed low VIF values (below 3), confirming they are not strongly collinear with each other. This finding is noted as a candidate refinement for future iterations of the model.

Chapter 7: Customer Churn Prediction Module

The final tuned Logistic Regression model was integrated into a prediction module that scores individual customers using both `predict()` for the churn classification and `predict_proba()` for the associated Stay Probability and Churn Probability.

7.1 Random Customer Prediction

A customer is randomly sampled from the feature matrix using `X.sample(1)`. The customer's full attribute profile is displayed, followed by the model's prediction and probability breakdown. In one such run, a customer with 7 months of tenure, Fiber-optic internet, no online security or tech support, a Month-to-month contract, and a Mailed check payment method was evaluated:

Attribute	Value
Tenure	7 months
Contract	Month-to-month
MonthlyCharges	₹74.65
TotalCharges	₹544.55
TotalServices	3
Predicted Outcome	Customer Will Churn
Stay Probability	46.16%
Churn Probability	53.84%

Figure 7.1: Random Customer Prediction Output

Business interpretation: with a Churn Probability of 53.84% (only marginally above the 50% decision threshold), this customer sits in a genuinely borderline risk zone. The short tenure, month-to-month commitment, and absence of value-added services (online security, tech support) are all consistent with the churn-risk factors identified during EDA and feature importance analysis, making this customer a reasonable candidate for a proactive retention outreach.

7.2 Manual Customer Scenario Prediction

In addition to random sampling, the module supports manually constructed customer profiles for what-if analysis. A hypothetical customer — male, with a partner, 5

months of tenure, Fiber-optic internet, six active services, a Month-to-month contract, paperless billing, and payment via Electronic check — was evaluated:

Attribute	Value
Tenure	5 months
Contract	Month-to-month
PaymentMethod	Electronic check
MonthlyCharges	₹95.60
TotalServices	6
HighValueCustomer	Yes
Predicted Outcome	Customer is likely to Churn
Churn Probability	68.42%

This scenario reflects a high-value, high-spend customer on a short, flexible contract — a profile the model flags with considerably higher churn risk (68.42%), consistent with the model's learned association between Electronic check payments, Month-to-month contracts, and elevated churn probability.

Chapter 8: Power BI Dashboard

To make the analytical findings and model outputs accessible to non-technical business stakeholders, the cleaned dataset and derived insights were imported into Microsoft Power BI. Power Query was used to shape the data, and a multi-page interactive dashboard was built to allow retention teams to explore churn drivers dynamically.

8.1 Executive Overview Dashboard

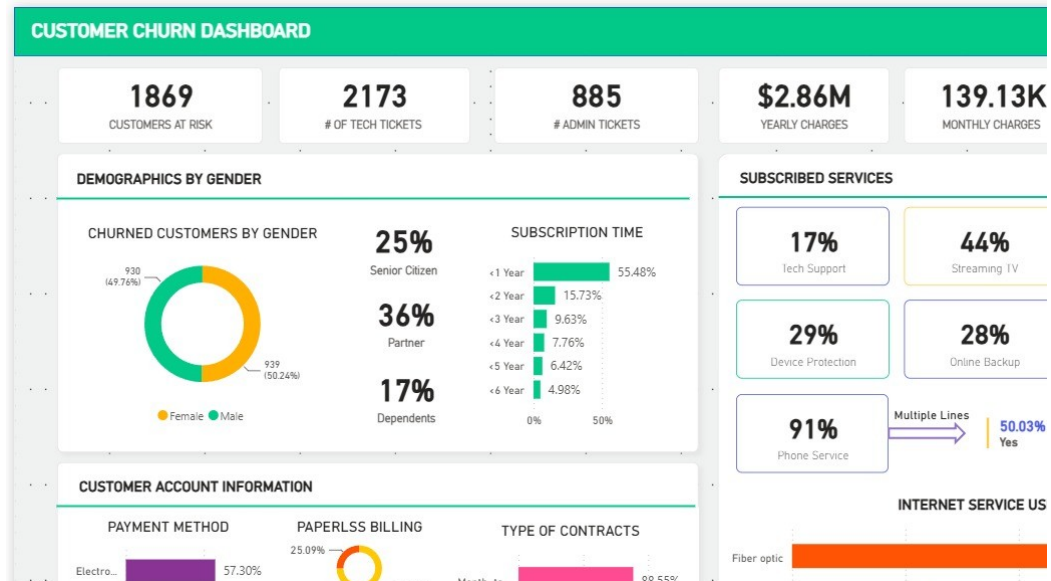


Figure 8.1: Power BI Dashboard Overview

This page presents high-level Key Performance Indicators — total customer count, overall churn rate, average tenure, and average monthly charges — giving stakeholders an immediate snapshot of the customer base and attrition scale. Slicers for Contract type, Internet service, and Payment method allow the KPIs to be filtered dynamically. The overview is designed as the entry point of the dashboard, orienting the viewer before they drill into specific churn drivers.

8.2 Customer Churn Insights Dashboard

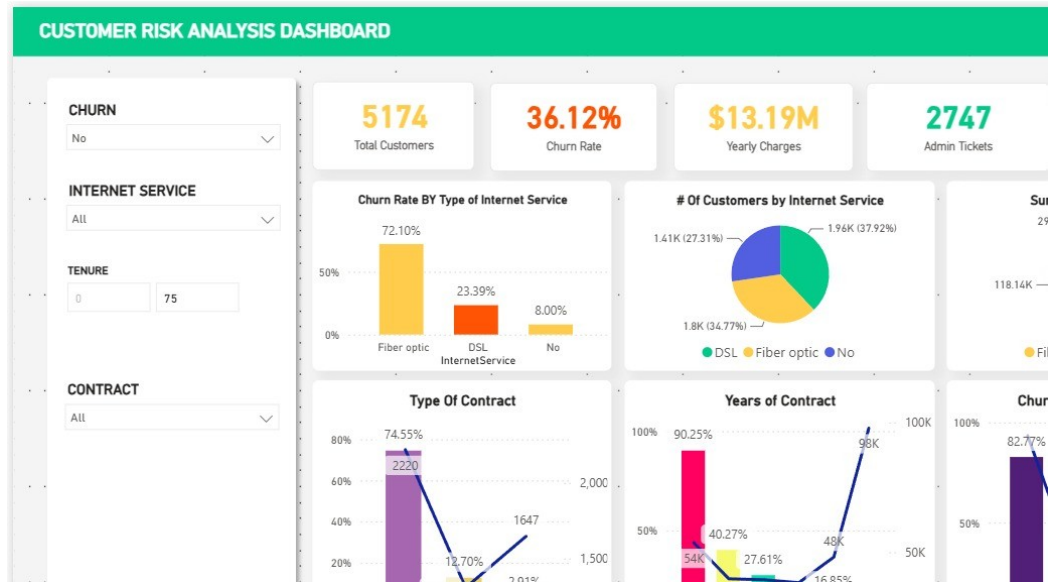
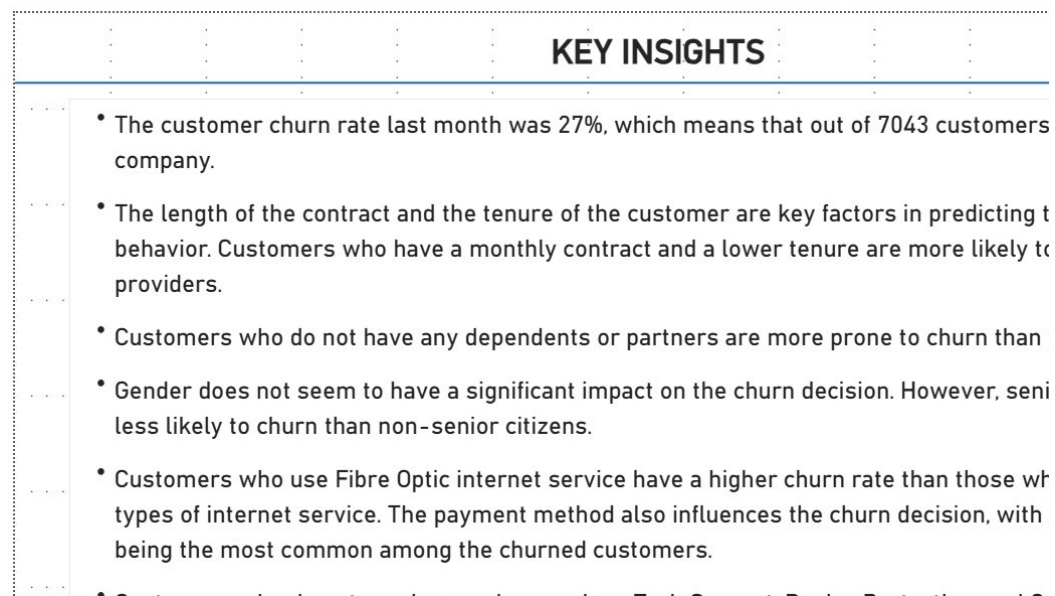


Figure 8.2: Customer Churn Insights Dashboard

This page visualises churn rate broken down by Contract type, Payment method, and Internet service, mirroring the bivariate EDA performed in the notebook. Stacked bar and donut charts highlight that Month-to-Month contracts and Electronic check payments are associated with the highest churn concentrations. Filters for tenure range and customer demographics let users isolate specific at-risk cohorts for targeted outreach planning.

8.3 Financial and Retention Dashboard



This page focuses on the financial dimension of churn, displaying average Monthly Charges and Total Charges by churn status, alongside CustomerLifetimeValue distributions for churned versus retained customers. It quantifies the revenue impact of churn and helps prioritise retention spend toward high-value, high-risk segments identified by the machine learning model.

Chapter 9: Results and Discussion

9.1 Summary of Findings

The exploratory phase established that churn is strongly associated with short tenure, Month-to-Month contracts, higher monthly charges, Electronic check payments, and the absence of value-added services such as Online Security and Tech Support. These findings were independently corroborated by the Logistic Regression feature-importance ranking and the statsmodels coefficient significance tests, giving confidence that the identified drivers are genuine rather than coincidental.

9.2 Model Performance

Among the three algorithms evaluated, Logistic Regression consistently outperformed Decision Tree and Random Forest on F1-Score and ROC-AUC, the metrics most relevant to an imbalanced classification problem. After hyperparameter tuning, the final model achieved 79.84% Accuracy, 0.638 Precision, 0.556 Recall, 0.594 F1-Score, and 0.721 ROC-AUC on unseen data — a dependable, interpretable baseline for churn scoring in a production retention workflow.

9.3 Business Value

Beyond raw predictive accuracy, the interpretability of Logistic Regression allowed each churn driver to be quantified and explained — a critical requirement for retention teams who must justify and design targeted interventions rather than act on an opaque "black box" score.

Chapter 10: Model Selection Rationale

Logistic Regression was selected as the final model over Decision Tree and Random Forest for several converging reasons, grounded directly in the comparative evaluation results.

- Best overall metrics: Logistic Regression achieved the highest Accuracy (0.796), F1-Score (0.585), and ROC-AUC (0.715) of the three baseline models, and remained the strongest performer after tuning.
- Better generalisation: the Decision Tree's comparatively weak test performance (Accuracy 0.720, ROC-AUC 0.649) relative to its likely training performance is consistent with overfitting on a fully-grown, unpruned tree.
- Balanced trade-off vs. Random Forest: Random Forest narrowed the Precision gap (0.621) but its Recall (0.487) and F1-Score (0.546) remained below Logistic Regression, meaning it missed more true churn cases despite its ensemble complexity.
- Interpretability: Logistic Regression coefficients directly quantify each feature's contribution and direction of influence, which was further validated statistically via statsmodels — an advantage Random Forest's less transparent feature importances do not offer to the same degree for business communication.
- Computational efficiency: Logistic Regression trains and scores near-instantly compared to a 200-tree Random Forest ensemble, an important consideration for real-time customer scoring.

Taken together, these factors made Logistic Regression the most defensible choice for a production-oriented churn scoring model, balancing predictive performance, statistical rigour, and business interpretability.

Chapter 11: Business Recommendations

Based on the exploratory findings and the model's feature importance results, the following retention strategies are recommended for telecom decision-makers:

- Encourage customers to move from Month-to-Month contracts to One-Year or Two-Year plans through pricing incentives or bundled discounts, since contract length is one of the strongest churn-reducing factors identified.
- Offer discounts and loyalty rewards to new customers during their first few months, as short-tenure customers show disproportionately higher churn.
- Promote automatic payment methods (bank transfer or credit card) over Electronic check, since automatic payments correlate with materially lower churn.
- Bundle or upsell Online Security and Tech Support add-ons, which the model identifies as churn-reducing services, potentially through free trial periods for at-risk segments.
- Prioritise retention outreach toward customers flagged as high monthly-charge, low-tenure, and Month-to-Month by the prediction model, since this profile combines high revenue value with elevated churn risk.
- Use the churn probability score to design a tiered retention programme — e.g., automated offers for medium-risk customers and personal account-manager contact for high-value, high-risk customers.

11.1 Business Perspective on Model Deployment

In practice, telecom companies can operationalise this model by scoring their active customer base on a recurring (e.g., monthly) basis. Customers crossing a defined churn-probability threshold can be automatically routed into a retention workflow — ranging from automated email offers to proactive calls from customer service — before they formally cancel. Because the model also exposes which factors are driving an individual customer's risk (e.g., contract type, payment method, missing add-on services), retention teams can personalise the specific offer rather than applying a generic discount, improving both retention rates and the cost-efficiency of retention spend.

Chapter 12: Conclusion and Future Scope

12.1 Conclusion

This project successfully delivered an end-to-end customer churn prediction system covering data cleaning, exploratory data analysis, feature engineering, model building, hyperparameter tuning, statistical validation, and model evaluation. Among Logistic Regression, Decision Tree, and Random Forest, Logistic Regression achieved the best overall performance and was selected as the final model, reaching an Accuracy of 79.84% and ROC-AUC of 0.721 after tuning. The integration of a random and manual prediction module, together with an interactive Power BI dashboard, translates the model's outputs into a practical decision-support tool for customer retention teams, while the accompanying business recommendations connect the analytical findings directly to actionable retention strategy.

12.2 Future Scope

- Address class imbalance using resampling techniques such as SMOTE, or algorithms with built-in class-weighting, to improve Recall on the minority Churn class.
- Reduce multicollinearity among engineered features (TotalServices, CustomerLifetimeValue, AvgMonthlySpend) identified through VIF analysis, potentially via feature selection or regularisation adjustments.
- Deploy the trained model as a REST API (e.g., using Flask or FastAPI) for real-time integration with customer relationship management (CRM) systems.
- Extend the Power BI dashboard with natural-language Q&A and automated threshold-based alerting for newly at-risk customers.
- Experiment with additional algorithms (e.g., Gradient Boosting, XGBoost) and probability calibration to further improve Recall without materially sacrificing Precision.

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